

Reminder: Monomial ideals are finitely generated

Initial ideal of $I = \langle \text{leading terms of polynomials in } I \rangle = \langle LT(I) \rangle$

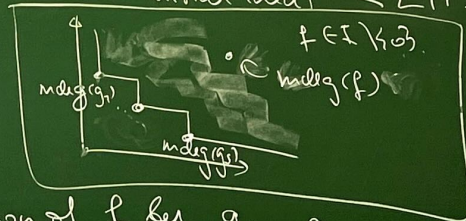
2.4.4 Theorem (Hilbert's basis theorem). Every ideal $I \subseteq k[x_1, \dots, x_n]$ is finitely generated.

Proof: We can assume $I \neq \{0\}$, as otherwise the assertion is clearly true. We pick finitely many polynomials $g_1, \dots, g_s \in I$, whose leading terms generate the initial ideal $\langle LT(I) \rangle$.

So, we have $\langle LT(g_1), \dots, LT(g_s) \rangle = \langle LT(I) \rangle$.

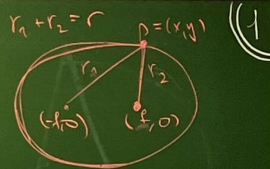
It turns out that $\langle g_1, \dots, g_s \rangle = I$.

To see this, we need to verify that every $f \in I$ belongs to $\langle g_1, \dots, g_s \rangle$.



obtaining $f = a_1 g_1 + \dots + a_s g_s + r$ with quotients $a_1, \dots, a_s \in k[x_1, \dots, x_n]$ and the remainder

$r \in k[x_1, \dots, x_n]$ where no term of r divides any of the leading terms of $g_1, \dots, g_s$. But $r = \underbrace{f}_{\in I} - \underbrace{a_1 g_1}_{\in I} - \dots - \underbrace{a_s g_s}_{\in I} \in I$. So, if we had $r \neq 0$, then $LT(r) \notin \langle LT(g_1), \dots, LT(g_s) \rangle$



and on the other hand $LT(r) \in \langle LT(I) \rangle$. But we've chosen $g_1, \dots, g_s$ such that $\langle LT(I) \rangle = \langle LT(g_1), \dots, LT(g_s) \rangle$.

So, we've got a contradiction showing that $r = 0$.

Consequently, $I = \langle g_1, \dots, g_s \rangle$. $\square$

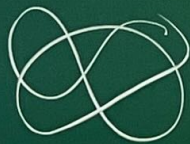
2.4.5 Theorem.

If $\langle LT(g_1), \dots, LT(g_s) \rangle = \langle LT(I) \rangle$, then $g_1, \dots, g_s$ is called a Gröbner basis (or a standard basis) of I . Of course, this depends on the underlying monomial order.

2.4.5' Rem.

Gröbner bases are a standard part of the computer algebra systems nowadays.

$$\begin{aligned}x &= t - t^2 + t^5 \\y &= t - 2t^3\end{aligned}$$



$$\langle x - t + t^2 - t^5, y - t + 2t^3 \rangle$$

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2.4.6 Corollary. Every ideal $I \subseteq k[x_1, x_n]$, $I \neq \{0\}$ has a Gröbner basis. A Gröbner basis of I is a basis of I .

Proof: see the proof of Hilbert's basis theorem. $\square$

2.4.7. Theorem. (ACC = the ascending chain condition). Consider an ^{infinite} sequence of ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ in $k[x_1, \dots, x_n]$. Then, there exists $N \in \mathbb{N}$ such that $I_N = I_{N+1} = I_{N+2} = \dots$ (this is called stabilization).

Proof: $I = \bigcup_{j=1}^{\infty} I_j$ is an ideal:

- $0 \in I_j$ for every $j \in \mathbb{N} \rightarrow 0 \in \bigcup_{j=1}^{\infty} I_j$
- $f \in \bigcup_{j=1}^{\infty} I_j, g \in \bigcup_{j=1}^{\infty} I_j \Rightarrow f \in I_s$ and $g \in I_t$ for some $s, t \in \mathbb{N} \Rightarrow f \in I_{\max\{s, t\}}, g \in I_{\max\{s, t\}} \Rightarrow f+g \in I_{\max\{s, t\}} \Rightarrow f+g \in \bigcup_{j=1}^{\infty} I_j$.
- $f \in k[x_1, \dots, x_n], g \in \bigcup_{j=1}^{\infty} I_j \Rightarrow f \in k[x_1, \dots, x_n]$ and $g \in I_s$ for some $s \in \mathbb{N} \Rightarrow f \cdot g \in I_s \Rightarrow f \cdot g \in \bigcup_{j=1}^{\infty} I_j$.

By Hilbert's basis theorem, I is finitely generated,

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that is, $I = \langle g_1, \dots, g_m \rangle$ for some $g_1, \dots, g_m \in I$.

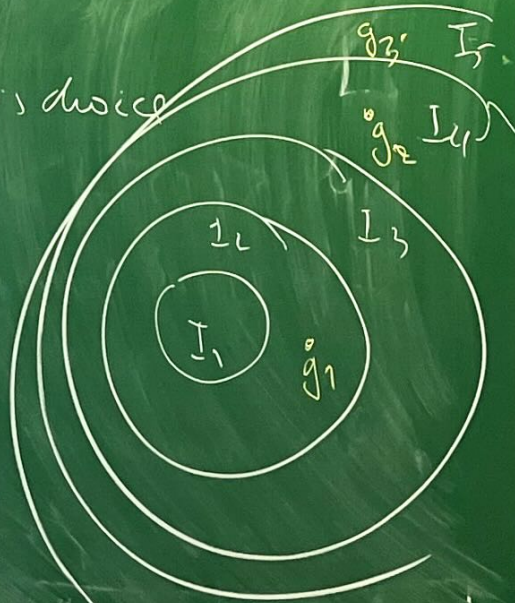
But $g_i \in \bigcup_{j=1}^{\infty} I_j$, so $g_i \in I_{s_i}$ for some $s_i \in \mathbb{N}$.

Let's take $N = \max\{s_1, \dots, s_m\}$. With this choice

so, $g_1, \dots, g_m \in I_j$ when $j \geq N_1 \Rightarrow$

$I = \langle g_1, \dots, g_m \rangle \subseteq I_j \subseteq I \Rightarrow$

$I = I_j$ for every $j \geq N_1$.



2.4.8. Def. Let $F \subseteq k[x_1, \dots, x_n]$, finite or infinite,

We define $V(F) := \{ a = (a_1, \dots, a_n) \in k^n : f(a_1, \dots, a_n) = 0 \text{ for all } f \in F \}$

(solution set of the system $f(x_1, \dots, x_n) = 0$ for $f \in F$, possibly infinite system).



$$f_5(x, y) = 0$$

2.4.9. Prop. For every $F \subseteq k[x_1, \dots, x_n]$ there exist finitely many polynomials $f_1, \dots, f_s \in F$ such that $V(F) = V(f_1, \dots, f_s)$.

Proof. Consider the ideal $I = \langle F \rangle$ generated by F . By Hilbert's Basis theorem,

$I = \langle g_1, \dots, g_m \rangle$ for some $g_1, \dots, g_m \in \langle F \rangle$. One has

$$g_1 = h_{1,1} f_{1,1} + \dots + h_{1,s_1} f_{1,s_1} \text{ with } h_{1,1}, \dots, h_{1,s_1} \in k[x_1, \dots, x_n] \text{ and } f_{1,1}, \dots, f_{1,s_1} \in F.$$

So, $g_1 \in \langle F_1 \rangle$ where $F_1 = \{f_{1,1}, \dots, f_{1,s_1}\}$ is a finite subset of F .

$\vdots$
 $g_m \in \langle F_m \rangle$ for some finite subset F_m of F .

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So, $I = \langle f_1, \dots, f_s \rangle$ when

$$\{f_1, \dots, f_s\} = F_1 \cup \dots \cup F_m.$$

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